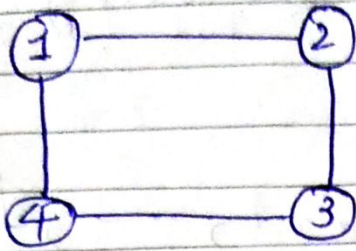


★ Hamiltonian Cycle starting



Point



1-2-3-4-1

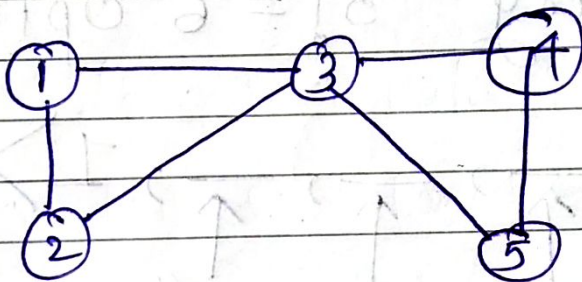


Ham. Cycle.

H.C.: Which has cycle in which every vertex appears exactly once (except starting vertex)

It is not always possible to have H.C. in the graph.

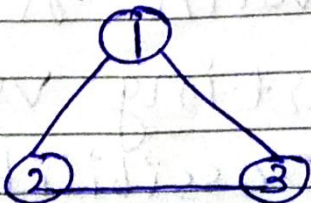
E.g.



There is no H.C. in the above graph irrespective of starting vertex because vertex 3 will come again in the cycle.

→ We can use all possible from the graph, provided it exist. Backtracking to find Hamiltonian Cycles

E.g.



1-2-3-1

1-3-2-1

2-3-1-2

2-1-3-2

3-1-2-3

3-2-1-3

→ We will assume that starting Vertex is 1.

$\langle 1, \text{---}, \text{---}, \text{---}, \dots, \text{---}, 1 \rangle$

Want to find such path, if possible.

→ If there are 4 vertices and if you use Brute force, then you need to try $3! = 6$ options

Cycle: $\langle 1, \text{---}, \text{---}, \text{---}, 1 \rangle$

$\uparrow$ $\uparrow$ $\uparrow$
 3 ways 2 ways 1 way

In Brute force I can generate every possibility and can see whether that possibility is valid or not.

So, for a graph with n vertices and assume starting vertex = 1, we have $(n-1)!$ possibilities to check.

For a graph with n vertices, the solution vector is represented as n -tuple:

$$\langle x[1], x[2], \dots, x[n] \rangle$$

If I start from vertex 1 then $x[1] = 1$

$$\langle 1, x[2], x[3], \dots, x[n] \rangle$$

Denotes
2nd vertex
in the
Hamiltonian
cycle

Constraints: There must be edge from $x[1]$ to $x[2]$

Similarly from $x[2]$ to $x[3]$
 $x[3]$ to $x[4]$

⋮

$x[n-1]$ to $x[n]$

$x[n]$ to $x[1]$

→ Each $x[i]$ is different
i.e. one vertex can't come
2 times in the solution.

→ $x[1] = 1$
 $x[2:n] = 0$

→ $\text{main}()$
{
 $H(2)$
}

→ $H(k)$
{
 do
 {
 $\text{NextValue}(k);$
 if ($x[k] == 0$)
 return;
 if ($k == n$)
 {
 $\text{print}(x[1:n]);$
 }
 else
 $H(k+1);$
 }
 while (true);
}

→ NextValue(K)

{
do

$x[K] = (x[K] + 1) \bmod (n + 1);$

if ($x[K] == 0$)
return;

if ($\kappa[x[K-1]][x[K]] \neq 0$)
{

for $j = 1$ to $K-1$

{

if ($x[j] == x[K]$)

{ break;

}

}

if ($j == K$)

{

if ($\kappa < n$ or ($\kappa == n$ and

$\kappa[x[n]][x[K]] \neq 0$))

{

return;

}

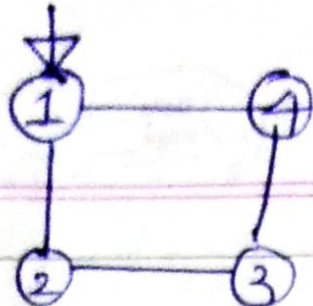
}

}

} while(true);

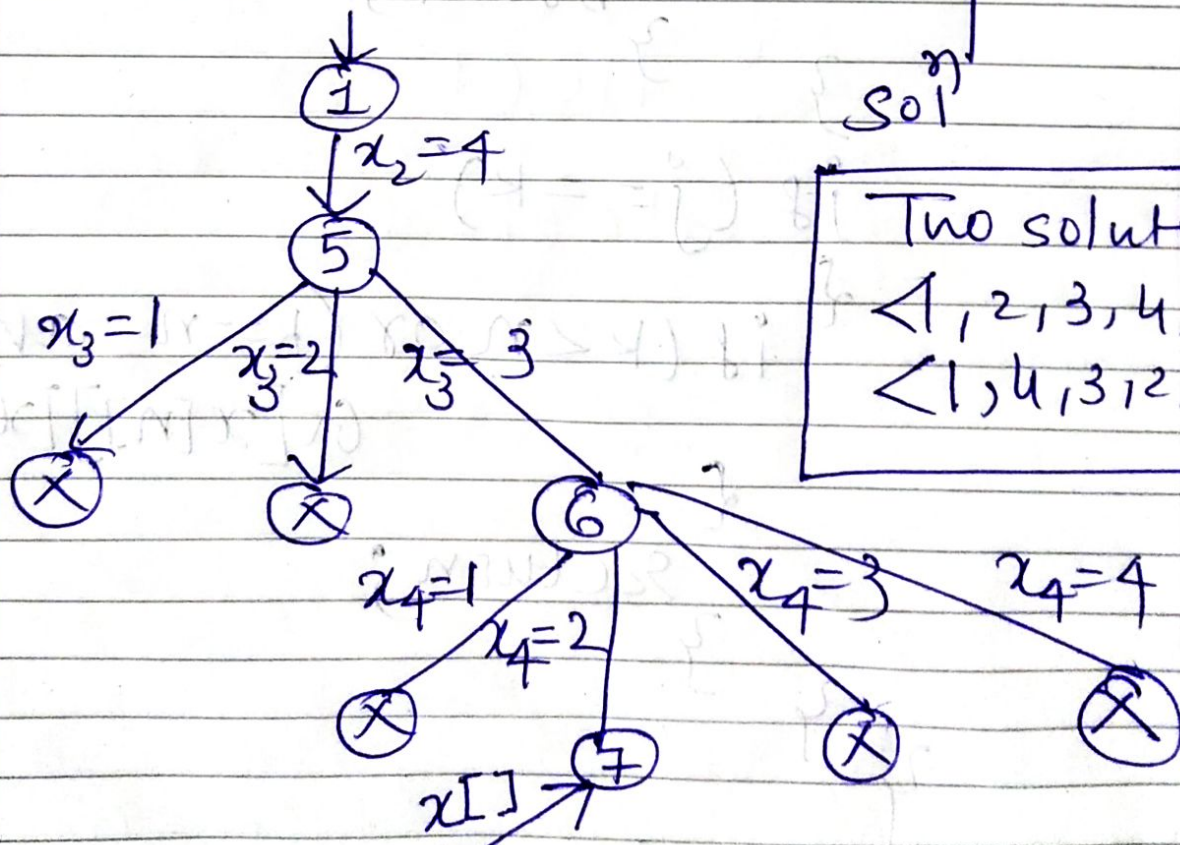
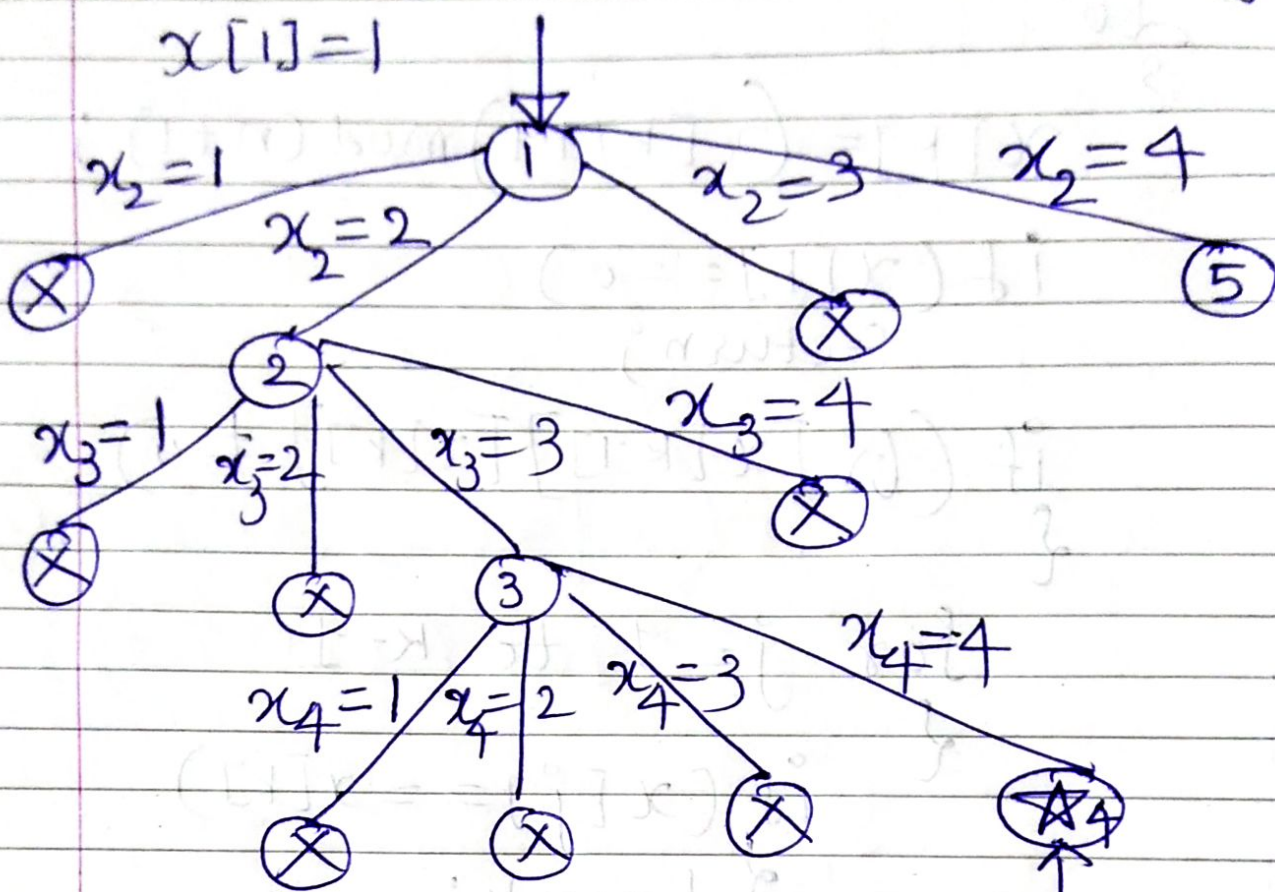
Graph

G:



$$x[1] = 1$$

Using Backtracking algo



Two solutions
 $\langle 1, 2, 3, 4, 1 \rangle$
 $\langle 1, 4, 3, 2, 1 \rangle$